

The difference between **subLVDS** and **subLVDS Emulation** on the Lattice CrossLink-NX FPGA lies in how the differential signaling is implemented, particularly in terms of hardware support and configuration. Below is a clear and concise explanation based on Lattice FPGA documentation and general FPGA I/O principles:

1. subLVDS:

- **Definition:** subLVDS (sub-Low-Voltage Differential Signaling) is a true differential signaling standard supported natively by the CrossLink-NX FPGA's dedicated differential I/O pairs in programmable I/O banks (e.g., Bank 1 and Bank 2). It operates with a VCCIO of 1.8V, a common-mode voltage of ~0.9V, and a differential output voltage (VOD) of ~150–300 mV.
- **Hardware:** Utilizes dedicated differential I/O pins with built-in differential drivers and receivers optimized for subLVDS signaling. These pins are designed to meet the electrical characteristics of the subLVDS standard, commonly used for low-power, low-voltage interfaces like those in image sensors (e.g., Sony camera modules).
- **Performance:** Provides optimal signal integrity and noise immunity for differential signaling, as it uses dedicated hardware designed for subLVDS. It supports high-speed data rates (up to 1.5 Gbps on CrossLink-NX, depending on configuration).
- **Use Case:** Ideal for direct connection to devices that natively output or accept subLVDS signals, such as certain image sensors, without requiring additional external components.

2. subLVDS Emulation:

- **Definition:** subLVDS Emulation refers to configuring single-ended I/O pins (typically LVCMOS18) to mimic subLVDS signaling by using two single-ended pins to emulate a differential pair. This is done through software configuration in the Lattice Radiant design tools, without relying on dedicated differential I/O hardware.
- **Hardware:** Uses general-purpose I/O pins rather than dedicated differential pairs. One pin is configured as the positive signal, and another as the negative (complementary) signal, with external resistors or termination to approximate subLVDS electrical characteristics (e.g., 100-ohm differential termination).
- **Performance:** Emulation may not achieve the same signal integrity, noise immunity, or speed as true subLVDS due to the lack of dedicated differential drivers/receivers. The maximum data rate may be lower, and performance depends on careful pin selection, board layout, and termination.
- **Use Case:** Employed when dedicated differential I/O pins are unavailable (e.g., already used for other interfaces) or when interfacing with a device that requires subLVDS-like signaling but can tolerate emulated characteristics. It's a workaround for flexibility in pin-constrained designs.

Key Differences:

Feature	subLVDS	subLVDS Emulation
I/O Type	Dedicated differential I/O pins	General-purpose single-ended I/O pins
Hardware	Uses built-in differential drivers	Emulates differential signaling via software
VCCIO	1.8V (native)	1.8V (configured as LVCMOS18)
Signal Integrity	High (optimized for subLVDS)	Lower (depends on board design)
Speed	Up to 1.5 Gbps (CrossLink-NX)	Typically lower, depends on configuration
Complexity	Simple (native support)	Requires careful pin assignment and termination
Use Case	Native subLVDS interfaces (e.g., image sensors)	Workaround when differential pins are unavailable

Relevance to HyperRAM:

As noted in prior responses, HyperRAM typically uses single-ended 1.8V LVCMOS signaling (e.g., LVCMOS18). Neither subLVDS nor subLVDS Emulation is directly compatible with HyperRAM's single-ended interface. However, **subLVDS Emulation** could theoretically be used to interface with a 1.8V device by emulating differential signaling, but this would require non-standard HyperRAM operation and external circuitry (e.g., differential-to-single-ended conversion), making LVCMOS18 the preferred choice for HyperRAM.

Verification:

- Check the **CrossLink-NX datasheet** or **Lattice Radiant software** for specific pin assignments and subLVDS/subLVDS Emulation support.
- Refer to the **Lattice FPGA I/O User Guide** for details on configuring emulated differential signaling.
- If considering subLVDS Emulation for a HyperRAM interface, validate the design with simulation or prototyping, as HyperRAM's single-ended LVCMOS18 is the standard approach.

For precise implementation details, consult Lattice Semiconductor's technical documentation or support team.